

Notes: Opp, Opp, and Harris (JFE, 2013)

Rating agencies in the face of regulation

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1 Big picture

- **Question.** How does rating-contingent regulation affect the informativeness of credit ratings and the volume of highly rated securities?
- **Core idea.** Ratings have two roles: they inform investors and they help regulated investors satisfy capital, collateral, or investment restrictions. When regulation gives highly rated securities an extra benefit, rating agencies may have incentives to inflate ratings rather than produce information.
- **Mechanism.** A rating agency can acquire costly information, truthfully disclose signals, and certify good issuers. But if the regulatory advantage of an A-rating is large enough, it can earn more by assigning A-ratings broadly and avoiding information-production costs.
- **Key threshold.** There is a threshold regulatory advantage $\bar{y}$. Below it, the agency acquires information and truthfully discloses signals. Above it, delegated information acquisition breaks down: all firms are rated A, disclosure is fully inflated, and information acquisition falls to zero.
- **Regulation result.** Rating-contingent regulation unambiguously raises the volume of highly rated securities. Its effect on informativeness can be positive or negative below the inflation threshold, but once the threshold is crossed, informativeness collapses.
- **Cross-asset-class result.** Rating inflation is more likely for complex securities, securities with high information-production costs, assets issued in boom times, asset classes with weak issuer outside options, and asset classes where regulation gives high ratings large benefits.
- **Why this is not just issuer-pays conflict.** Investors are not fooled in the model. Rating inflation can arise because regulated investors value the regulatory benefit of A-ratings even when everyone understands the rating agency's incentives.
- **Connection to the crisis.** The theory helps explain why rating standards deteriorated more for some structured products than for corporate bonds: regulation made the high-rating label valuable, and complexity made costly information acquisition less attractive.
- **Policy implication.** Reducing the regulatory use of ratings, as mandated by Dodd-Frank, can reduce the incentive to inflate ratings, but the effect may differ across asset classes.

2 Setup and measurement in the theory

2.1 Agents and projects

- There is a continuum of firms of mass one.
- Each firm has one project and no internal cash.
- The project type is $n \in \{g, b\}$: good (g) or bad (b).
- A good project has positive NPV, while a bad project has negative NPV.
- The share of good projects is π_g , and the share of bad projects is $\pi_b = 1 - \pi_g$.
- Firms know their own type, investors do not, and the rating agency can produce a noisy signal about type.

Interpretation: The asymmetric-information problem is adverse selection. Without sufficiently informative certification, public debt markets cannot separate positive-NPV from negative-NPV issuers.

2.2 Rating agency technology

The rating agency chooses costly information production $\iota \in [0, 1/2]$. Its private signal is $s \in \{A, B\}$, where

$$\Pr(s = A \mid n = g) = \Pr(s = B \mid n = b) = 1 - \alpha(\iota), \quad (1)$$

with

$$\alpha(\iota) = \frac{1}{2} - \iota. \quad (2)$$

Information costs satisfy

$$C'(0) = 0, \quad \lim_{\iota \rightarrow 1/2} C'(\iota) = \infty, \quad (3)$$

and C is increasing and convex.

Interpretation: Higher ι means a more accurate signal. The convex cost function captures the idea that rating more complex securities precisely is expensive.

2.3 Rating disclosure rule

The agency can issue an indicative rating $\tilde{r} \in \{A, B\}$, which becomes public only if the issuer purchases the rating. A key disclosure parameter is ε :

- if $s = A$, the agency offers $\tilde{r} = A$;
- if $s = B$, the agency offers $\tilde{r} = A$ with probability ε ;
- if $\varepsilon = 0$, disclosure is truthful;
- if $\varepsilon = 1$, all firms receiving public ratings can become A-rated.

Interpretation: The agency can distort disclosure because its private signal is not publicly verifiable. Rating inflation means upward bias from bad signals to A-ratings.

2.4 Rating-contingent regulation

Let $y \geq 0$ denote the regulatory advantage of an A-rated security. Economically, y summarizes lower capital requirements, wider investment eligibility, better collateral treatment, or any other regulatory benefit attached to a high rating.

Interpretation: The paper takes y as an exogenous positive variable and studies how it changes private certification incentives. The main comparative-static object is how ι^* , ε^* , and the mass of A-rated securities change with y .

2.5 Outside options

Good firms have an outside option $\bar{U}_g$, while bad firms have $\bar{U}_b = 0$. The outside option can be interpreted as another certification technology, a bank relationship, private information production, or an alternative way to signal quality.

Interpretation: A stronger outside option limits the rents the rating agency can extract from good firms. This affects whether truthful information production or rating inflation is more profitable.

3 Model equations and equilibrium logic

3.1 Posterior quality of A-rated securities

Let $\mu_A(\iota)$ and $\mu_B(\iota)$ be the masses of firms receiving signals A and B :

$$\mu_A(\iota) = \pi_g[1 - \alpha(\iota)] + \pi_b\alpha(\iota), \quad (4)$$

$$\mu_B(\iota) = \pi_g\alpha(\iota) + \pi_b[1 - \alpha(\iota)]. \quad (5)$$

Given disclosure rule ε , the mass of firms with an indicative A-rating is

$$\tilde{\mu}_A(\iota, \varepsilon) = \mu_A(\iota) + \varepsilon\mu_B(\iota). \quad (6)$$

The posterior default probability of a public A-rating is

$$d_A(\iota, \varepsilon) = \frac{\pi_g[1 - \alpha(\iota)(1 - \varepsilon)]}{\tilde{\mu}_A}d_g + \frac{\pi_b[\alpha(\iota) + (1 - \alpha(\iota))\varepsilon]}{\tilde{\mu}_A}d_b. \quad (7)$$

Interpretation: More information production improves the quality of A-rated securities under truthful disclosure. More inflation raises the mass of A-rated securities but lowers their average quality.

3.2 Investor break-even condition

Investors provide financing if the face value N_A needed to break even is no greater than the maximum pledgeable amount $\bar{N}$. The break-even condition can be written as

$$N_A(\iota, \varepsilon, f, y) = \frac{1 + f - y}{1 - d_A(\iota, \varepsilon)}. \quad (8)$$

Therefore the maximum fee the agency can charge is

$$f \leq f^*(\iota, \varepsilon, y) = [1 - d_A(\iota, \varepsilon)]\bar{N} + y - 1. \quad (9)$$

Interpretation: The regulatory advantage y directly relaxes the financing constraint and raises the fee the agency can charge for an A-rating.

3.3 Revenue contribution of type n

The paper defines the contribution from an A-rated type- n firm as

$$x_n(y) = (1 - d_n)\bar{N} + y - 1. \quad (10)$$

Because bad projects have negative NPV, $x_b(0) < 0$. A regulatory advantage can make high ratings more profitable by raising $x_g(y)$ and $x_b(y)$.

Interpretation: The central tension is that identifying bad firms saves losses, but rating all firms A creates more fee-paying volume and allows all issuers to obtain regulatory benefits.

3.4 Rating agency problem

The rating agency chooses information production ι , disclosure ε , and fee f :

$$\max_{\iota, \varepsilon, f} \Pi(\iota, \varepsilon, f, y) = \tilde{\mu}_A(\iota, \varepsilon) f - C(\iota) \quad (11)$$

subject to the financing constraint

$$N_A(\iota, \varepsilon, f, y) \leq \bar{N}. \quad (12)$$

Since the fee constraint binds, the agency's problem reduces to choosing information quality and disclosure to maximize rating revenue net of information costs.

Interpretation: The agency internalizes both the price of a high rating and the number of issuers that can be sold one. This is why regulation affects both information production and disclosure standards.

3.5 Benchmark without rating-contingent regulation

When $y = 0$, the equilibrium has truthful disclosure ($\varepsilon^* = 0$). The optimal information level satisfies

$$C'(\iota^*) = \pi_g x_g(0) - \pi_b x_b(0). \quad (13)$$

Interpretation: Without regulatory advantage, the agency maximizes profits by separating good projects from bad projects. Since bad projects are costly to finance, the agency has incentives to acquire information.

3.6 Rating inflation threshold

For $y > 0$, there is a unique threshold $\bar{y} \in (0, |x_b(0)|)$ such that

- if $y \leq \bar{y}$, full disclosure remains optimal and $\iota^*(y) > 0$;
- if $y > \bar{y}$, all firms are rated A, $\varepsilon = 1$, and no information is produced, $\iota = 0$.

The threshold is implicitly defined by equality between the full-disclosure profit and the rating-inflation profit.

Interpretation: The discontinuity is the key result. Once the regulatory benefit becomes large enough, the agency does not merely reduce standards gradually. It may switch to pure regulatory arbitrage.

3.7 Comparative statics

The threshold $\bar{y}$ is:

- decreasing in information-cost parameters;
- increasing in good issuers' outside option $\bar{U}_g$;
- decreasing in the payoff from success R ;
- decreasing in the fraction of good types π_g .

Interpretation: Rating inflation is more likely for complex assets, in asset classes where good issuers have weak alternatives, and in boom states where the agency can sell many A-ratings.

4 Identification and empirical design implications

4.1 What kind of evidence the theory points to

The paper is theoretical, so there is no direct empirical design in the main model. But the theory produces observable comparative statics:

- asset classes with stronger rating-contingent regulatory benefits should have more highly rated securities;
- informativeness can decline sharply when the regulatory advantage crosses a threshold;
- complex assets should be more vulnerable to inflation because information acquisition is costly;
- rating inflation should be more likely in boom times when the fraction of good issuers or project payoffs are high;
- regulatory reforms that reduce reliance on ratings should reduce inflation incentives.

4.2 Why investors do not need to be fooled

A major point of the paper is that rating inflation can occur even when investors understand the rating agency's incentives. Investors may still value A-rated securities because regulation makes A-ratings valuable.

Interpretation: The model shifts the explanation from naive investors to regulatory arbitrage. The A-rating is valuable partly because regulation says it is valuable.

4.3 Why asset complexity matters

Information production costs $C(\iota)$ determine the profitability of truthful certification. More complex securities raise these costs, which lowers the inflation threshold $\bar{y}$.

Interpretation: This helps explain why structured products were more vulnerable to rating inflation than corporate bonds. Corporate bonds were more familiar and cheaper to evaluate.

4.4 Dodd-Frank implications

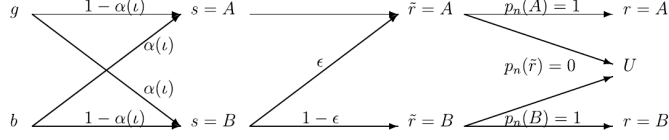
The Dodd-Frank Act mandated reductions in the use of ratings in regulation. In the model, this means lowering y . Lower y reduces the profitability of pure rating inflation and can move an asset class back below the threshold where information production is sustainable.

Interpretation: The effect is not guaranteed to be uniform. Asset classes with high information costs or weak outside options may remain vulnerable even when regulatory reliance is reduced.

5 Figures and original-paper excerpts

The paper has three main figures and several formal propositions. The visuals and theorem excerpts below are directly cropped from the original paper and grouped compactly.

5.1 Figure 1 and signal equations: rating process and information technology



(a) Fig. 1: signal, indicative rating, and public rating.

signal, we consider the following signal structure (see the left panel of Fig. 1):

$$\Pr(s=A|n=g) = \Pr(s=B|n=b) = 1-\alpha(i), \quad (2)$$

where $i \in [0, \frac{1}{2}]$ denotes the rating agency's choice of information production. Importantly, the quality of the rating agency's signal, $1-\alpha(i)$, is endogenous. Signals are informative if the error probability $\alpha(i)$ is smaller than 50%.²² It is convenient and without loss of generality to assume α is affine; that is,

$$\alpha(i) = \frac{1}{2} - i. \quad (3)$$

(b) Signal structure and information-cost assumptions.

Read:

- The rating agency sees a private binary signal about issuer type.
- It can upward-bias disclosure by offering an A-rating even after a B-signal.
- Issuers choose whether to buy and publish the rating.
- Only purchased ratings become public; otherwise the firm remains unrated.

Economic message: The rating agency controls both information production and disclosure. This is the source of the trade-off between certification and regulatory arbitrage.

5.2 Proposition 1 and Proposition 2: benchmark and threshold

the case of $y > 0$.

Proposition 1. The benchmark equilibrium is characterized by

- (a) full disclosure ($\varepsilon^* = 0$),
- (b) the level of information acquisition satisfies $C'(i^*) = \pi_g x_g(0) - \pi_b x_b(0)$,
- (c) the fee satisfies $f^*(i^*, 0, 0) = \bar{N}(1 - d_A(i^*, 0)) - 1$,
- (d) the fraction of firms financed through the bond market is $\mu_A(i^*)$, and
- (e) rating agency profits are given by $(1 - \alpha(i^*))\pi_g x_g(0) + \alpha(i^*)\pi_b x_b(0) - C(i^*)$.

Proof. See Appendix B and Online Appendix.

(a) Proposition 1: benchmark equilibrium.

we now present the main result of this subsection and one of the main results of the paper.

Proposition 2. There exists a unique threshold level of the regulatory advantage, $\bar{y} \in (0, |x_b(0)|)$, such that full disclosure of information $i^(y)$ is optimal if $y \leq \bar{y}$. Otherwise, all firms are rated A ($\varepsilon = 1$) and no information ($i = 0$) is produced. The threshold level of the regulatory advantage is defined implicitly by the equation*

$$(1 - \alpha(i^*(\bar{y})))\pi_g x_g(\bar{y}) + \alpha(i^*(\bar{y}))\pi_b x_b(\bar{y}) - C(i^*(\bar{y})) = \pi_g x_g(\bar{y}) + \pi_b x_b(\bar{y}), \quad (16)$$

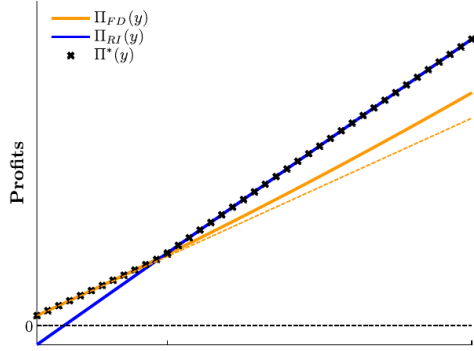
(b) Proposition 2: unique inflation threshold.

Read:

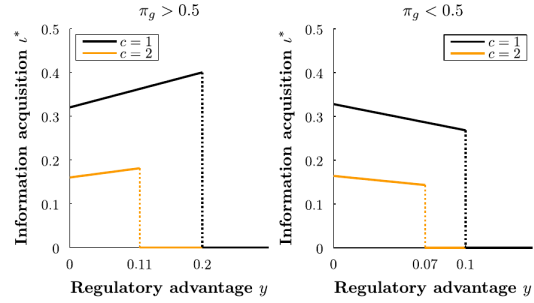
- Proposition 1 states that without regulatory advantage ($y = 0$), the agency truthfully discloses information and chooses an interior signal precision.
- Proposition 2 states that a unique threshold $\bar{y}$ separates full disclosure from full rating inflation.
- Above the threshold, the agency sets $\varepsilon = 1$ and $i = 0$: all firms are rated A and no information is produced.

Economic message: The model does not predict merely continuous deterioration. It predicts the possibility of a regime switch from certification to regulatory arbitrage.

5.3 Figure 2 and Figure 3: profit comparison and comparative statics



(a) Fig. 2: full disclosure versus rating inflation profits.



(b) Fig. 3: information acquisition as regulatory advantage rises.

Read:

- Figure 2 plots profits under full disclosure and rating inflation as functions of the regulatory advantage.
- At the threshold, the two profit functions cross.
- Figure 3 shows that information production can rise or fall with y below the threshold, depending on the share of good types.
- Once the threshold is crossed, information acquisition jumps to zero.

Economic message: Regulation always increases the value of A-ratings, but it does not always increase rating informativeness. The threshold effect is central.

5.4 Proposition 3 and Corollaries: who is vulnerable to rating inflation?

Determinants of information provision in the full-disclosure region: In the following, we analyze the determinants of information provision i in the full-disclosure region ($y \leq \bar{y}$). First, consider the effect of the regulatory advantage y on the amount of information acquisition i^* and the mass of A-rated firms μ_A .³⁵

Proposition 3. *In the full-disclosure region ($y \leq \bar{y}$), an increase in y increases information acquisition if and only if $\pi_g > \frac{1}{2}$. Otherwise, information acquisition decreases. The mass of A-rated firms strictly increases with increases in y for*

(a) Proposition 3: full-disclosure region.

satisfies $\pi_g = 0.7 > \frac{1}{2}$. In contrast, the right panel plots a case in which $\pi_g = 0.2 < \frac{1}{2}$ so that information acquisition decreases. The “volume channel” of regulation, the incentive to label more firms as A in response to a preferential regulatory treatment of A-rated securities, drives these comparative statics of informativeness. Since in the full-disclosure region, the fraction of A-rated firms is increasing in i if and only if $\pi_g > \frac{1}{2}$, the sign of the “volume effect” on equilibrium information acquisition depends solely on the proportions of the two types π_n .

Corollary 2. *In the full-disclosure region ($y < \bar{y}$), information acquisition is*

- (1) *decreasing in the cost parameter c and independent of k ,*
- (2) *increasing in $\bar{U}_g$ for $\pi_g < \kappa/(1+\kappa)$ and decreasing otherwise, where $\kappa \equiv (1-d_b)/(1-d_g) < 1$ represents the ratio of success probabilities for the two firm types,*
- (3) *decreasing in R for $\pi_g < \kappa/(1+\kappa)$ and increasing otherwise,*
- (4) *increasing in π_g for $x_g(y) + x_b(y) > 0$ and decreasing*

(c) Corollary 2: information acquisition comparative statics.

standards when the regulatory advantage of highly rated securities is below the threshold $\bar{y}$. In addition to the effect of y , we consider comparative statics with respect to

1. the parameters c and k for the class of cost functions $C_{c,k}(i) = cC(i) + k$, where $c, k \in \mathbb{R}^+$,³³
2. the outside option of good issuers $\bar{U}_g$,
3. the payoff for success R , and

(b) Corollary 1: determinants of the threshold.

Read:

- Proposition 3 explains how information provision changes below the inflation threshold.
- If the fraction of good types is high, more regulatory advantage can increase information production because the agency gains more from identifying good firms.
- If the fraction of good types is low, more regulatory advantage can reduce information production.
- Corollary 1 identifies forces that lower the inflation threshold: higher information costs, weaker outside options, higher payoff from success, and a higher fraction of good types.

Economic message: The model predicts heterogeneous rating standards across asset classes and over the credit cycle.

6 Short summary

- The paper studies credit rating agencies when ratings are used both for information and for regulation.
- Rating-contingent regulation gives A-rated securities an extra benefit y .
- A rating agency chooses information production, disclosure, and fees.
- Without regulation, the agency truthfully discloses information and acquires costly signals.

- With small regulatory advantages, information production can rise or fall depending on asset-class characteristics.
- With sufficiently large regulatory advantage, the agency prefers rating inflation: all issuers are rated A and no information is produced.
- The model predicts more highly rated securities whenever ratings confer regulatory benefits.
- Rating inflation is more likely when information costs are high, good issuers' outside options are weak, project payoffs are high, or the fraction of good issuers is high.
- Investors do not need to be irrational. They may knowingly accept inflated ratings because regulation gives high ratings private value.
- The framework explains variation in rating standards across asset classes, especially between structured products and corporate bonds.

7 Conclusion

7.1 Core conclusion

Opp, Opp, and Harris show that rating-contingent regulation can undermine the informational role of ratings. The same regulatory rules that make highly rated securities attractive to institutional investors can make it privately optimal for rating agencies to inflate ratings.

7.2 Economic intuition

The rating agency earns fees from issuers that can obtain financing. Information production helps identify good issuers, but it is costly. Regulation increases the value of an A-rating, so the agency can sometimes earn more by expanding the A-rated pool than by screening carefully. When the regulatory advantage is high enough, the agency switches from certification to regulatory arbitrage.

7.3 Takeaway

The paper's main insight is that rating failures can arise from the interaction of private certification and public regulation, not only from naive investors or the issuer-pays model. Rating-contingent regulation may increase the quantity of highly rated securities while reducing the quality of the information those ratings provide.